

Database System Engineering and Distributed Backend Development

Course Code: 25CS13002E

ROLL NO:2520030414

Name: NAGABHYRAVA YISHNAANI

Query 1: Create the database

```
CREATE DATABASE IF NOT EXISTS ddl_db;  
USE ddl_db;
```

Output: -

✓	39	15:09:29	CREATE DATABASE IF NOT EXISTS ddl_db	1 row(s) affected	0.110 sec
✓	40	15:09:34	USE ddl_db	0 row(s) affected	0.015 sec

Query 2: -DDL (DATA DEFINITION LANGUAGE)

CREATE TABLE

```
CREATE TABLE bank_transactions (  
    txn_id INT PRIMARY KEY,  
    customer_name VARCHAR(50),  
    branch_name VARCHAR(50),  
    transaction_type VARCHAR(20),  
    amount DECIMAL(10,2),  
    transaction_date DATE  
);
```

Output: -

✓	41	15:12:39	CREATE TABLE bank_transactions (txn_id INT P...	0 row(s) affected	0.203 sec
---	----	----------	--	-------------------	-----------

Query 3: -ALTER TABLE ADD COLUMN

```
);  
ALTER TABLE bank_transactions  
ADD account_no VARCHAR(20);
```

Output: -

✓	42	15:14:01	ALTER TABLE bank_transactions ADD account_no ...	0 row(s) affected Records: 0 Duplicates: 0 Warnings: 0	0.093 sec
---	----	----------	--	--	-----------

Query 4: -ALTER TABLE MODIFY COLUMN

```
ALTER TABLE bank_transactions  
MODIFY customer_name VARCHAR(100);
```

Output: -

✓	43	15:15:05	ALTER TABLE bank_transactions MODIFY customer...	0 row(s) affected	Records: 0 Duplicates: 0 Warnings: 0	0.109 sec
---	----	----------	--	-------------------	--------------------------------------	-----------

Query 5: -RENAME TABLE

```
RENAME TABLE  
bank_transactions TO customer_transactions;
```

Output: -

✓	44	15:17:35	RENAME TABLE bank_transactions TO customer_tr...	0 row(s) affected		0.063 sec
---	----	----------	--	-------------------	--	-----------

Query 6: -TRUNCATE TABLE

```
TRUNCATE TABLE customer_transactions;
```

Output: -

✓	46	15:19:47	TRUNCATE TABLE customer_transactions	0 row(s) affected		0.032 sec
---	----	----------	--------------------------------------	-------------------	--	-----------

Query 7: -DROP TABLE

```
DROP TABLE IF EXISTS bank_backup;
```

Output: -

⚠	48	15:21:33	DROP TABLE IF EXISTS bank_backup	0 row(s) affected, 1 warning(s): 1051 Unknown table '...		0.109 sec
---	----	----------	----------------------------------	--	--	-----------

Query 8: - DML (DATA MANIPULATION LANGUAGE)

INSERT SAMPLE DATA

```

1 CREATE TABLE customers_transactions (
2     txn_id INT PRIMARY KEY,
3     customer_name VARCHAR(100),
4     branch_name VARCHAR(50),
5     transaction_type VARCHAR(20),
6     amount DECIMAL(10,2),
7     transaction_date DATE
8 );

INSERT INTO customers_transactions VALUES
(101, 'YISHAANI', 'Hyderabad', 'DEPOSIT', 25000, '2026-01-15'),
(102, 'RAHUL', 'Mumbai', 'WITHDRAWAL', 5000, '2026-01-18'),
(103, 'PRIYA', 'Delhi', 'DEPOSIT', 18000, '2026-02-02'),
(104, 'ARJUN', 'Bangalore', 'WITHDRAWAL', 7500, '2026-02-10'),
(105, 'ANANYA', 'Chennai', 'DEPOSIT', 32000, '2026-02-14'),
(106, 'ROHAN', 'Pune', 'WITHDRAWAL', 4200, '2026-02-20'),
(107, 'SAI', 'Hyderabad', 'DEPOSIT', 15000, '2026-03-01'),
(108, 'NEHA', 'Kolkata', 'WITHDRAWAL', 6800, '2026-03-05'),
(109, 'VIKRAM', 'Jaipur', 'DEPOSIT', 27500, '2026-03-12'),
(110, 'KAVYA', 'Vijayawada', 'WITHDRAWAL', 3500, '2026-03-18'),
(111, 'ADITYA', 'Delhi', 'DEPOSIT', 22000, '2026-04-01'),
(112, 'SNEHA', 'Mumbai', 'WITHDRAWAL', 9000, '2026-04-07'),
(113, 'KARTHIK', 'Chennai', 'DEPOSIT', 12500, '2026-04-15'),
(114, 'MEGHA', 'Pune', 'WITHDRAWAL', 5600, '2026-04-22'),
(120, 'TANVI', 'Delhi', 'WITHDRAWAL', 6300, '2026-06-25');

```

Output: -

✓	52	15:27:20	CREATE TABLE customers_transactions (txn_id I...	0 row(s) affected	0.031 sec
✓	53	15:27:28	INSERT INTO customers_transactions VALUES (101...	10 row(s) affected Records: 10 Duplicates: 0 Wamin...	0.015 sec

Query 9: -INSERT

```

INSERT INTO customers_transactions
VALUES
(111, 'Venu', 'Vizag', 'Deposit', 7000, '2024-01-18');

```

Output: -

✓	54	15:28:57	INSERT INTO customers_transactions VALUES (111...	1 row(s) affected	0.062 sec
---	----	----------	---	-------------------	-----------

Query 10: -UPDATE

```

UPDATE customers_transactions
SET amount = 5000
WHERE txn_id = 105;

```

Output: -

56 15:30:53 UPDATE customers_transactions SET amount = 500... 1 row(s) affected Rows matched: 1 Changed: 1 War... 0.015 sec

Query 11: -DELETE

```
DELETE FROM customers_transactions
WHERE txn_id = 111;
```

Output: -

58 15:32:09 DELETE FROM customers_transactions WHERE txn... 1 row(s) affected 0.000 sec

Query 12: -SELECT

```
SELECT *
FROM customers_transactions;
```

Output: -

60 15:33:32 SELECT * FROM customers_transactions LIMIT 0, 10... 10 row(s) returned 0.016 sec / 0.000 sec

104	Anirudh Reddy	Vizag	Deposit	8000.00	2024-01-10	20
105	Yashwanth	Hyderabad	Withdrawal	5000.00	2024-01-11	
106	Adithya Vardhan	Vizag	Deposit	15000.00	2024-01-12	
107	Ananaya Varshitha	Vijayawada	Withdrawal	1000.00	2024-01-13	
108	Varnika Guptha	Hyderabad	Deposit	9000.00	2024-01-14	
109	Purna	Vizag	Withdrawal	4000.00	2024-01-15	
110	Nihita	Vijayawada	Deposit	11000.00	2024-01-16	
*	NULL	NULL	NULL	NULL	NULL	

Query 13: -SELECT WITH WHERE

```
SELECT *
FROM customers_transactions
WHERE transaction_type='Deposit';
```

Output: -

62 15:35:53 SELECT * FROM customers_transactions WHERE tra... 6 row(s) returned 0.000 sec / 0.000 sec

104	Anirudh Reddy	Vizag	Deposit	8000.00	2024-01-10	
106	Adithya Vardhan	Vizag	Deposit	15000.00	2024-01-12	
108	Varnika Guptha	Hyderabad	Deposit	9000.00	2024-01-14	
110	Nihita	Vijayawada	Deposit	11000.00	2024-01-16	
*	NULL	NULL	NULL	NULL	NULL	

Query 14: -SELECT WITH ORDER BY

```
SELECT *  
FROM customers_transactions  
ORDER BY amount DESC;
```

Output: -

64 15:37:59 SELECT * FROM customers_transactions ORDE... 10 row(s) returned 0.016 sec / 0.000 sec

Result Grid

Filter Rows:

Edit:

Export/Import:

	txn_id	customer_name	branch_name	transaction_type	amount	transaction_date
▶	106	Adithya Vardhan	Vizag	Deposit	15000.00	2024-01-12
	103	Akshara Reddy	Vijayawada	Deposit	12000.00	2024-01-08
	110	Nihita	Vijayawada	Deposit	11000.00	2024-01-16
	108	Varnika Guptha	Hyderabad	Deposit	9000.00	2024-01-14

Query 15: - DCL (DATA CONTROL LANGUAGE)

GRANT ALL

```
GRANT ALL PRIVILEGES  
ON ddl_db.customers_transactions  
TO 'BranchManager'@'localhost';  
SHOW GRANTS FOR 'BranchManager'@'localhost';
```

Output: -

72 15:48:57 GRANT ALL PRIVILEGES ON ddl_db.customers... 0 row(s) affected 0.000 sec

Query 16: -REVOKE ALL

```
REVOKE ALL PRIVILEGES  
ON ddl_db.customers_transactions  
FROM 'BranchManager'@'localhost';
```

Output: -

74 15:49:33 REVOKE ALL PRIVILEGES ON ddl_db.customer... 0 row(s) affected 0.000 sec

Query 17: -TCL (TRANSACTION CONTROL LANGUAGE)

COMMIT

```
UPDATE customer_transactions  
SET amount=6000  
WHERE txn_id=101;  
  
COMMIT;
```

Output: -

✓	76	15:52:18	UPDATE customer_transactions SET amount=60...	0 row(s) affected Rows matched: 0 Changed: 0 ...	0.000 sec
✓	77	15:52:25	COMMIT	0 row(s) affected	0.000 sec

Query 18: - SAVEPOINT

```
SAVEPOINT Before_Update;
```

Output: -

✓	78	15:53:32	SAVEPOINT Before_Update	0 row(s) affected	0.000 sec
---	----	----------	-------------------------	-------------------	-----------

Query 19: -ROLLBACK TO SAVEPOINT

```
UPDATE customer_transactions  
SET amount=99999  
WHERE txn_id=102;  
ROLLBACK TO Before_Update;
```

Output: -

✓	84	15:55:57	UPDATE customer_transactions SET amount=99...	0 row(s) affected Rows matched: 0 Changed: 0 ...	0.000 sec
---	----	----------	---	--	-----------